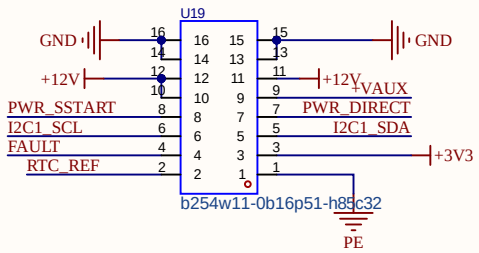
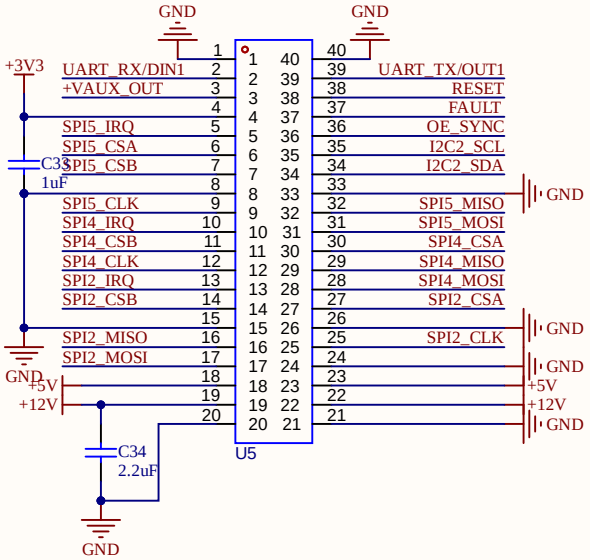
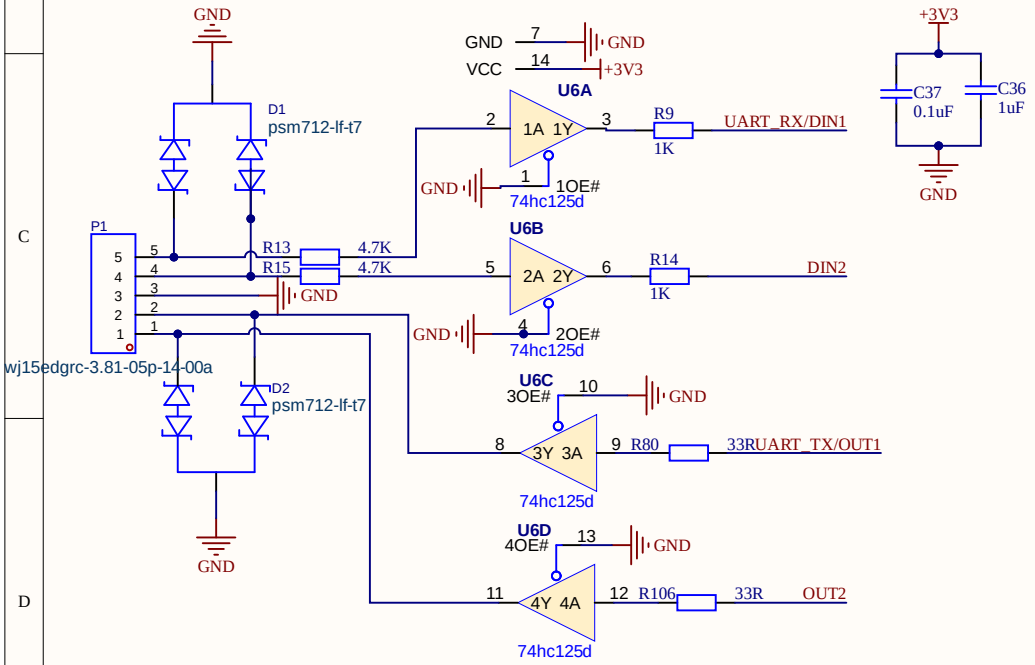
## 输入接口（电源、软启动、风扇控制器）



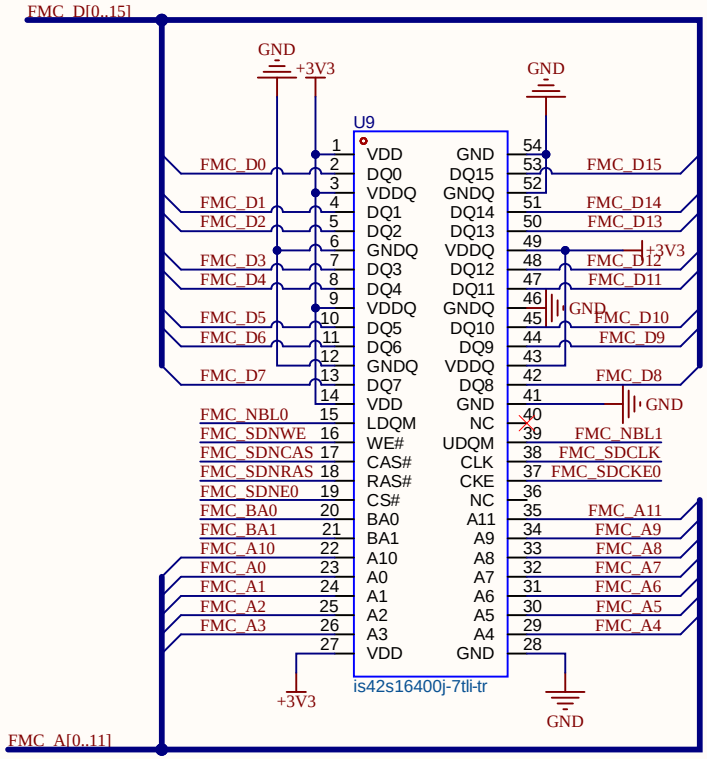

## 输出连接器（3 个 SPI 模块）

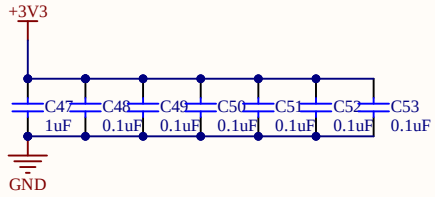



## 带保护的数字输入输出

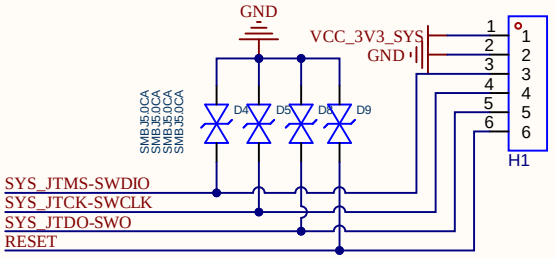


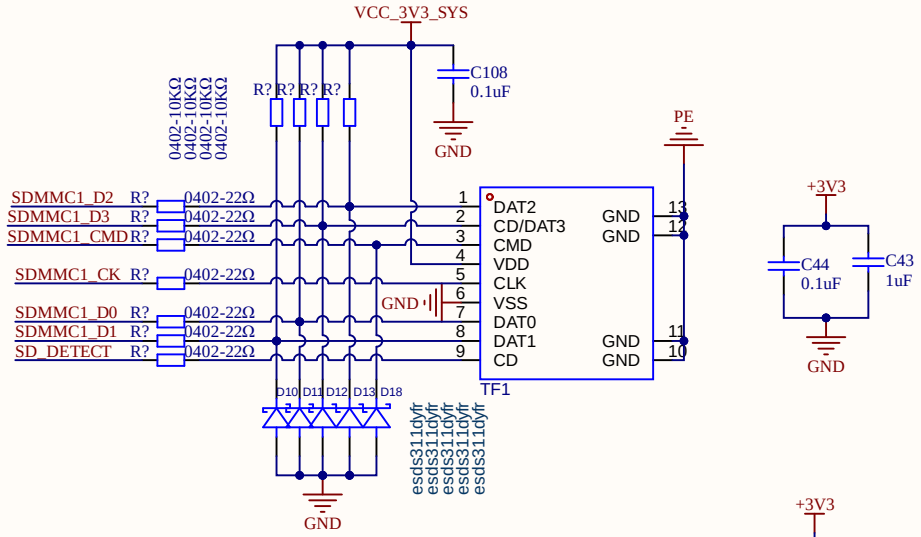

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| Date: 10/28/2026 | | Sheet of |
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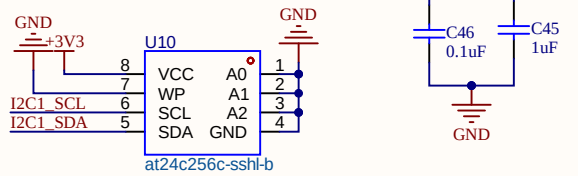


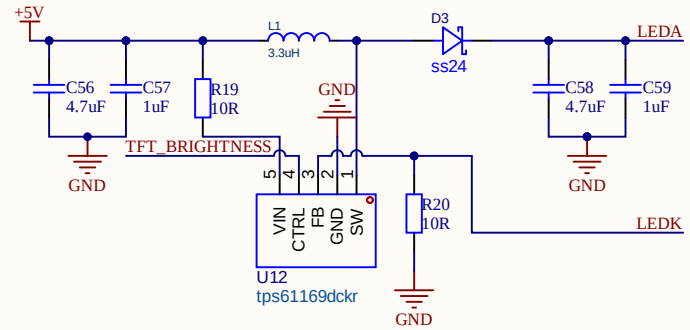

# JTAG/SWD

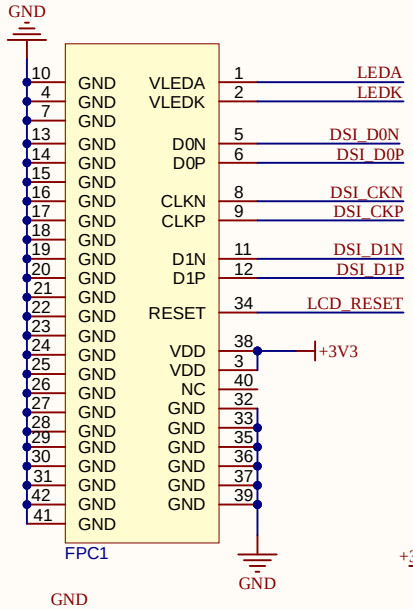


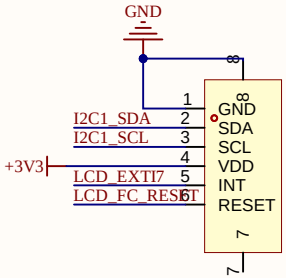



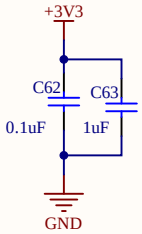




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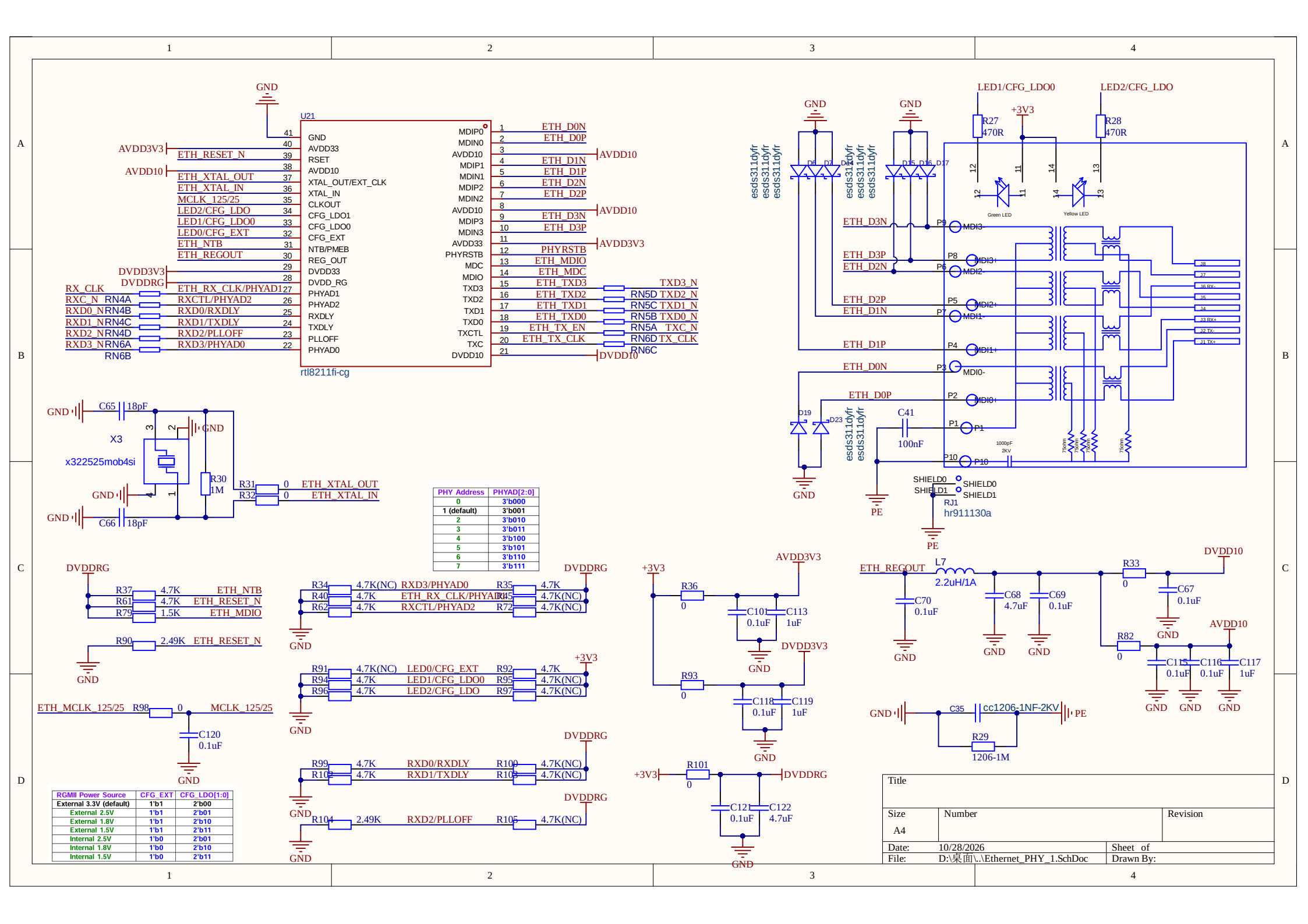


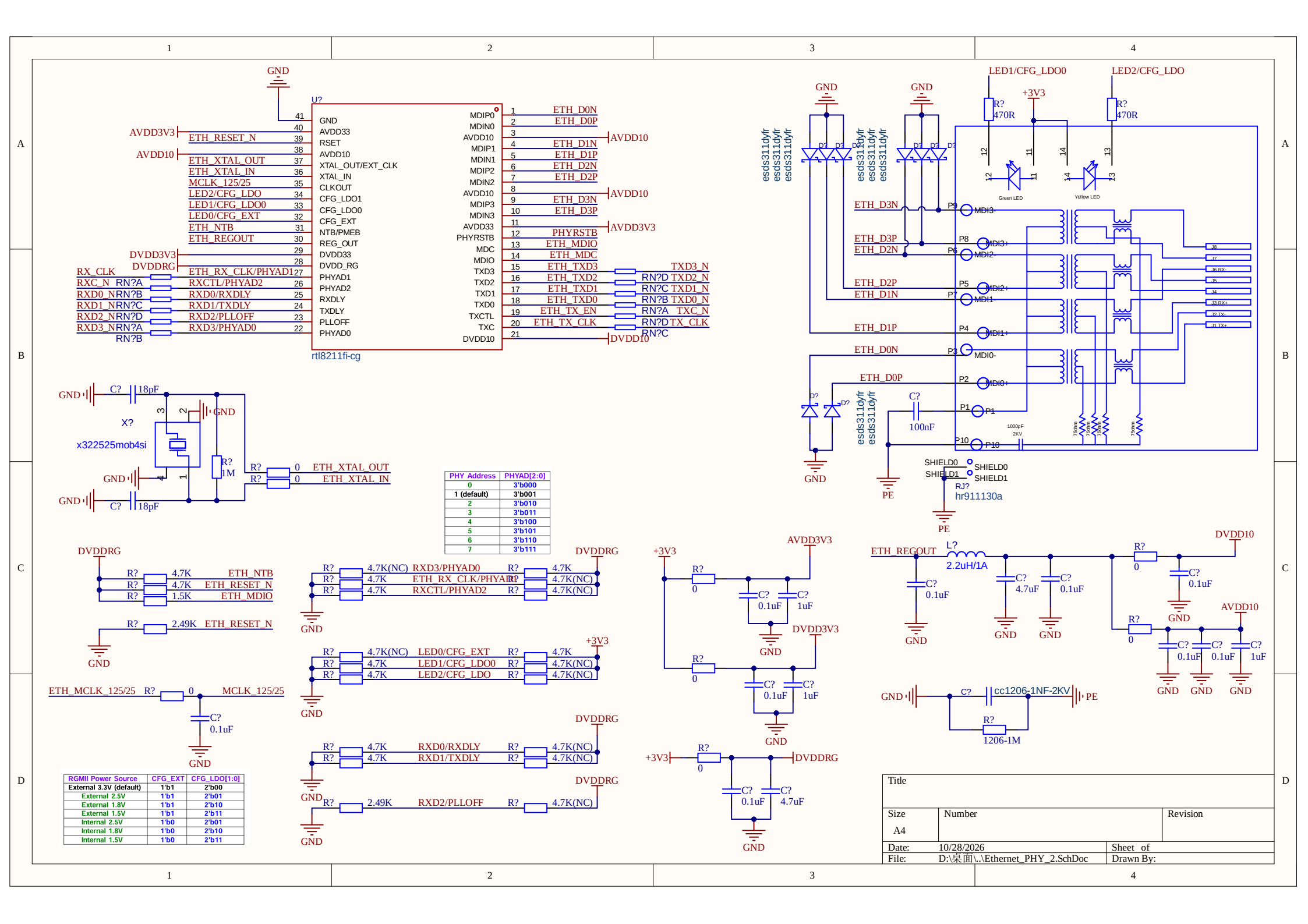



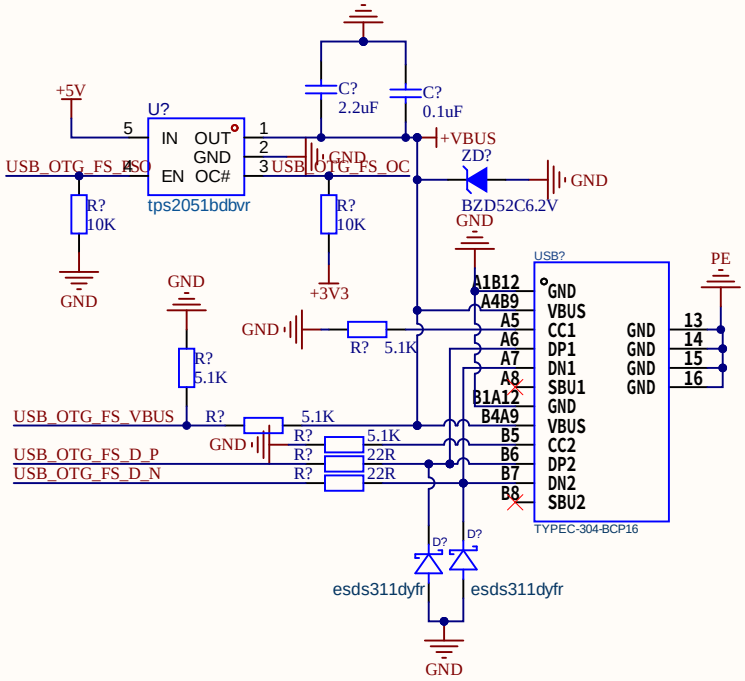



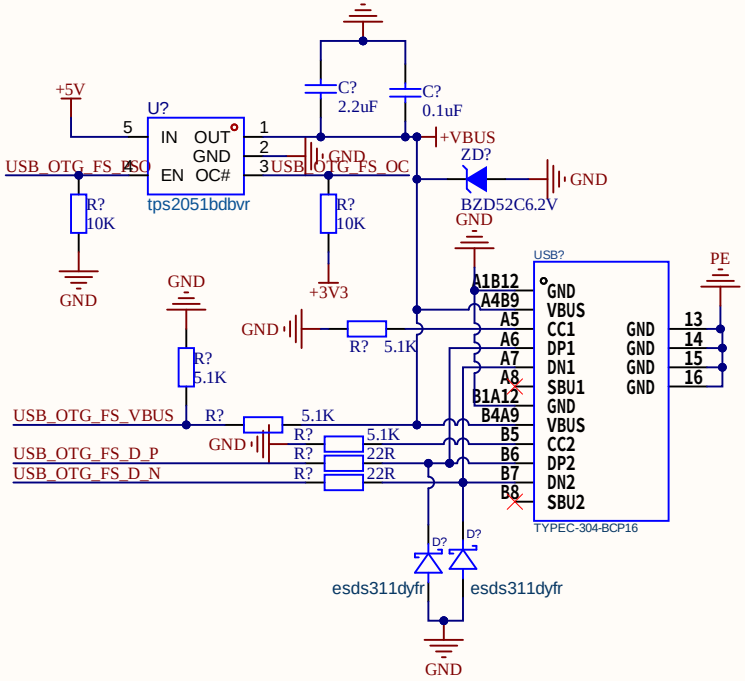




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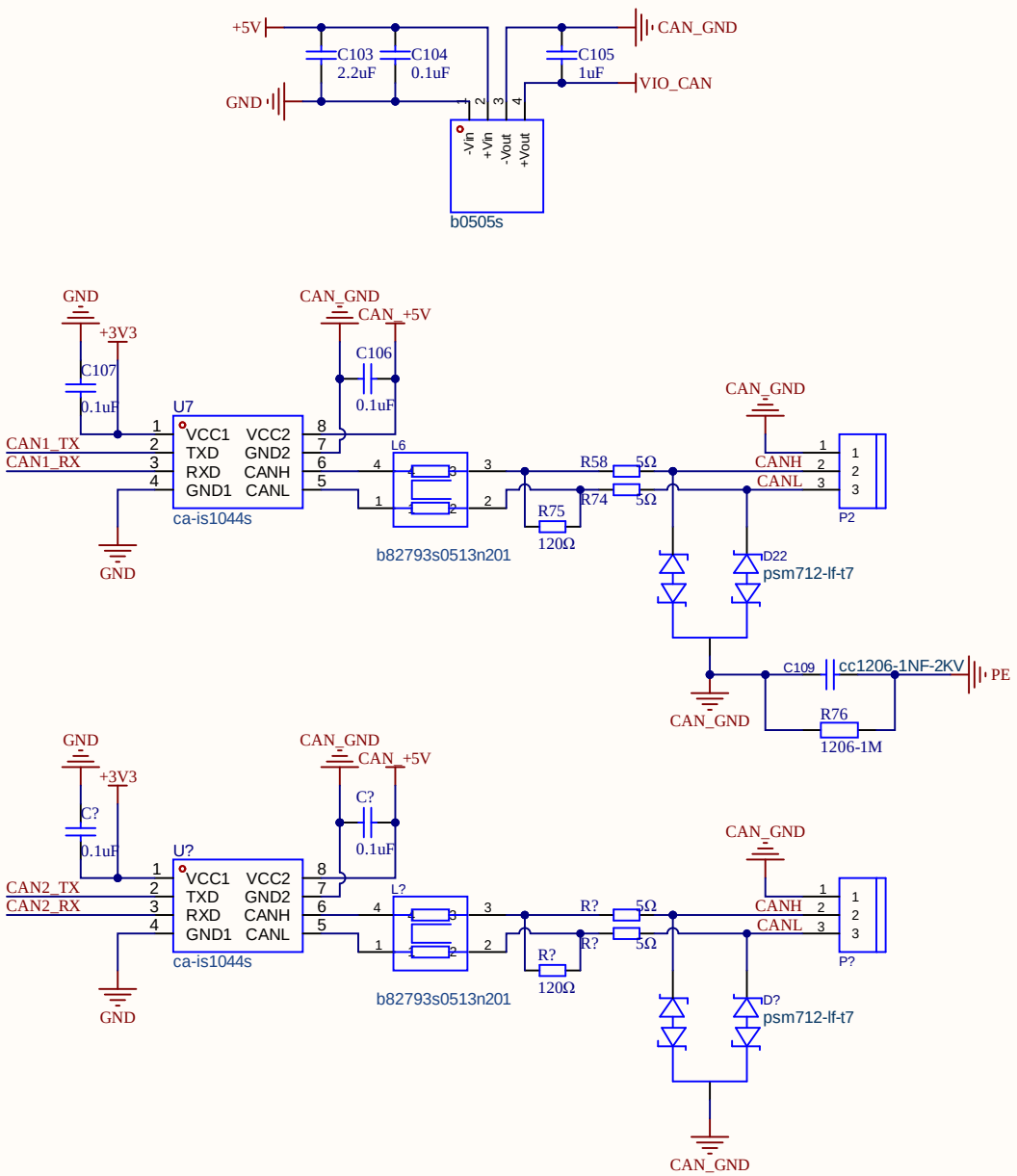

U21
rtl8211fi-cg
GND
AVDD3V3
AVDD10
DVDD3V3
DVDDRG
ETH_RESET_N
ETH_XTAL_OUT
ETH_XTAL_IN
MCLK_125/25
LED2/CFG_LDO
LED1/CFG_LDO0
LED0/CFG_EXT
ETH_NTB
ETH_REGOUT
ETH_RX_CLK/PHYAD1
RXCTL/PHYAD2
RXD0/RXDLY
RXD1/TXDLY
RXD2/PLLOFF
RXD3/PHYAD0
RN6B
RN4A
RN4B
RN4C
RN4D
RN6A
RX_CLK
RXC_N
RXD0_N
RXD1_N
RXD2_N
RXD3_N
GND
AVDD33
RSET
AVDD10
XTAL_OUT/EXT_CLK
XTAL_IN
CLKOUT
CFG_LDO1
CFG_LDO0
CFG_EXT
NTB/PMEB
REG_OUT
DVDD33
DVDD_RG
PHYAD1
PHYAD2
RXDLY
TXDLY
PLLOFF
PHYAD0
MDIP0
MDIN0
AVDD10
MDIP1
MDIN1
MDIP2
MDIN2
AVDD10
MDIP3
MDIN3
AVDD33
PHYRSTB
MDC
MDIO
TXD3
TXD2
TXD1
TXD0
TXCTL
TXC
DVDD10
ETH_D0N
ETH_D0P
ETH_D1N
ETH_D1P
ETH_D2N
ETH_D2P
ETH_D3N
ETH_D3P
PHYRSTB
ETH_MDIO
ETH_MDC
ETH_TXD3
ETH_TXD2
ETH_TXD1
ETH_TXD0
ETH_TX_EN
ETH_TX_CLK
TXD3_N
RN5D TXD2_N
RN5C TXD1_N
RN5B TXD0_N
RN5A TXC_N
RN6D TX_CLK
RN6C
X3
x322525mob4si
C65 18pF
C66 18pF
R30 1M
R31 0 ETH_XTAL_OUT
R32 0 ETH_XTAL_IN
PHY Address | PHYAD[2:0]
0 | 3'b000
1 (default) | 3'b001
2 | 3'b010
3 | 3'b011
4 | 3'b100
5 | 3'b101
6 | 3'b110
7 | 3'b111
R37 4.7K ETH_NTB
R61 4.7K ETH_RESET_N
R79 1.5K ETH_MDIO
R90 2.49K ETH_RESET_N
R34 4.7K(NC) RXD3/PHYAD0
R40 4.7K ETH_RX_CLK/PHYAD1
R62 4.7K RXCTL/PHYAD2
R35 4.7K
R45 4.7K(NC)
R72 4.7K(NC)
R91 4.7K(NC) LED0/CFG_EXT
R94 4.7K LED1/CFG_LDO0
R96 4.7K LED2/CFG_LDO
R92 4.7K
R95 4.7K(NC)
R97 4.7K(NC)
ETH_MCLK_125/25 R98 0 MCLK_125/25
C120 0.1uF
R99 4.7K RXD0/RXDLY
R102 4.7K RXD1/TXDLY
R100 4.7K(NC)
R103 4.7K(NC)
R104 2.49K RXD2/PLLOFF
R105 4.7K(NC)
RGMII Power Source | CFG_EXT | CFG_LDO[1:0]
External 3.3V (default) | 1'b1 | 2'b00
External 2.5V | 1'b1 | 2'b01
External 1.8V | 1'b1 | 2'b10
External 1.5V | 1'b1 | 2'b11
Internal 2.5V | 1'b0 | 2'b01
Internal 1.8V | 1'b0 | 2'b10
Internal 1.5V | 1'b0 | 2'b11
R36 0
C101 0.1uF
C113 1uF
R93 0
C118 0.1uF
C119 1uF
R101 0
C121 0.1uF
C122 4.7uF
LED1/CFG_LDO0
LED2/CFG_LDO
R27 470R
R28 470R
+3V3
Green LED
Yellow LED
esds311dyfr
D6 D7 D8 D15 D16 D17 D19 D23
ETH_D3N ETH_D3P ETH_D2N ETH_D2P ETH_D1N ETH_D1P ETH_D0N ETH_D0P
P9 MDI3-
P8 MDI3+
P6 MDI2-
P5 MDI2+
P7 MDI1-
P4 MDI1+
P3 MDI0-
P2 MDI0+
P1 P1
P10 P10
C41 100nF
1000pF 2KV
75ohm
SHIELD0
SHIELD1
RJ1
hr911130a
PE
L7 2.2uH/1A
C70 0.1uF
C68 4.7uF
C69 0.1uF
R33 0
C67 0.1uF
R82 0
C115 0.1uF
C116 0.1uF
C117 1uF
C35 cc1206-1NF-2KV
R29 1206-1M
Title
Size A4
Number
Revision
Date: 10/28/2026
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U?
rtl8211fi-cg
GND
AVDD3V3
AVDD10
DVDD3V3
DVDDRG
ETH_RESET_N
ETH_XTAL_OUT
ETH_XTAL_IN
MCLK_125/25
LED2/CFG_LDO
LED1/CFG_LDO0
LED0/CFG_EXT
ETH_NTB
ETH_REGOUT
RX_CLK
RXC_N RN?A
RXD0_NRN?B
RXD1_NRN?C
RXD2_NRN?D
RXD3_NRN?A
RN?B
ETH_RX_CLK/PHYAD1
RXCTL/PHYAD2
RXD0/RXDLY
RXD1/TXDLY
RXD2/PLLOFF
RXD3/PHYAD0
GND
AVDD33
RSET
AVDD10
XTAL_OUT/EXT_CLK
XTAL_IN
CLKOUT
CFG_LDO1
CFG_LDO0
CFG_EXT
NTB/PMEB
REG_OUT
DVDD33
DVDD_RG
PHYAD1
PHYAD2
RXDLY
TXDLY
PLLOFF
PHYAD0
MDIP0
MDIN0
AVDD10
MDIP1
MDIN1
MDIP2
MDIN2
AVDD10
MDIP3
MDIN3
AVDD33
PHYRSTB
MDC
MDIO
TXD3
TXD2
TXD1
TXD0
TXCTL
TXC
DVDD10
ETH_D0N
ETH_D0P
ETH_D1N
ETH_D1P
ETH_D2N
ETH_D2P
ETH_D3N
ETH_D3P
PHYRSTB
ETH_MDIO
ETH_MDC
ETH_TXD3
ETH_TXD2
ETH_TXD1
ETH_TXD0
ETH_TX_EN
ETH_TX_CLK
TXD3_N
RN?D TXD2_N
RN?C TXD1_N
RN?B TXD0_N
RN?A TXC_N
RN?D TX_CLK
RN?C
X?
x322525mob4si
C? 18pF
R? 1M
R? 0 ETH_XTAL_OUT
R? 0 ETH_XTAL_IN
PHY Address | PHYAD[2:0]
0 | 3'b000
1 (default) | 3'b001
2 | 3'b010
3 | 3'b011
4 | 3'b100
5 | 3'b101
6 | 3'b110
7 | 3'b111
RGMII Power Source | CFG_EXT | CFG_LDO[1:0]
External 3.3V (default) | 1'b1 | 2'b00
External 2.5V | 1'b1 | 2'b01
External 1.8V | 1'b1 | 2'b10
External 1.5V | 1'b1 | 2'b11
Internal 2.5V | 1'b0 | 2'b01
Internal 1.8V | 1'b0 | 2'b10
Internal 1.5V | 1'b0 | 2'b11
4.7K ETH_NTB
4.7K ETH_RESET_N
1.5K ETH_MDIO
2.49K ETH_RESET_N
4.7K(NC) RXD3/PHYAD0
4.7K ETH_RX_CLK/PHYAD1
4.7K RXCTL/PHYAD2
4.7K(NC) LED0/CFG_EXT
4.7K(NC) LED1/CFG_LDO0
4.7K(NC) LED2/CFG_LDO
4.7K(NC) RXD0/RXDLY
4.7K(NC) RXD1/TXDLY
2.49K RXD2/PLLOFF
4.7K(NC)
ETH_MCLK_125/25 R? 0 MCLK_125/25
C? 0.1uF
LED1/CFG_LDO0
LED2/CFG_LDO
+3V3
R? 470R
Green LED
Yellow LED
esds311dyfr
ETH_D3N
ETH_D3P
ETH_D2N
ETH_D2P
ETH_D1N
ETH_D1P
ETH_D0N
ETH_D0P
MDI3-
MDI3+
MDI2-
MDI2+
MDI1-
MDI1+
MDI0-
MDI0+
P1
P10
C? 100nF
1000pF 2KV
SHIELD0
SHIELD1
RJ?
hr911130a
PE
ETH_REGOUT
L? 2.2uH/1A
C? 0.1uF
C? 4.7uF
C? 1uF
cc1206-1NF-2KV
R? 1206-1M
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Size A4
Number
Revision
Date: 10/28/2026
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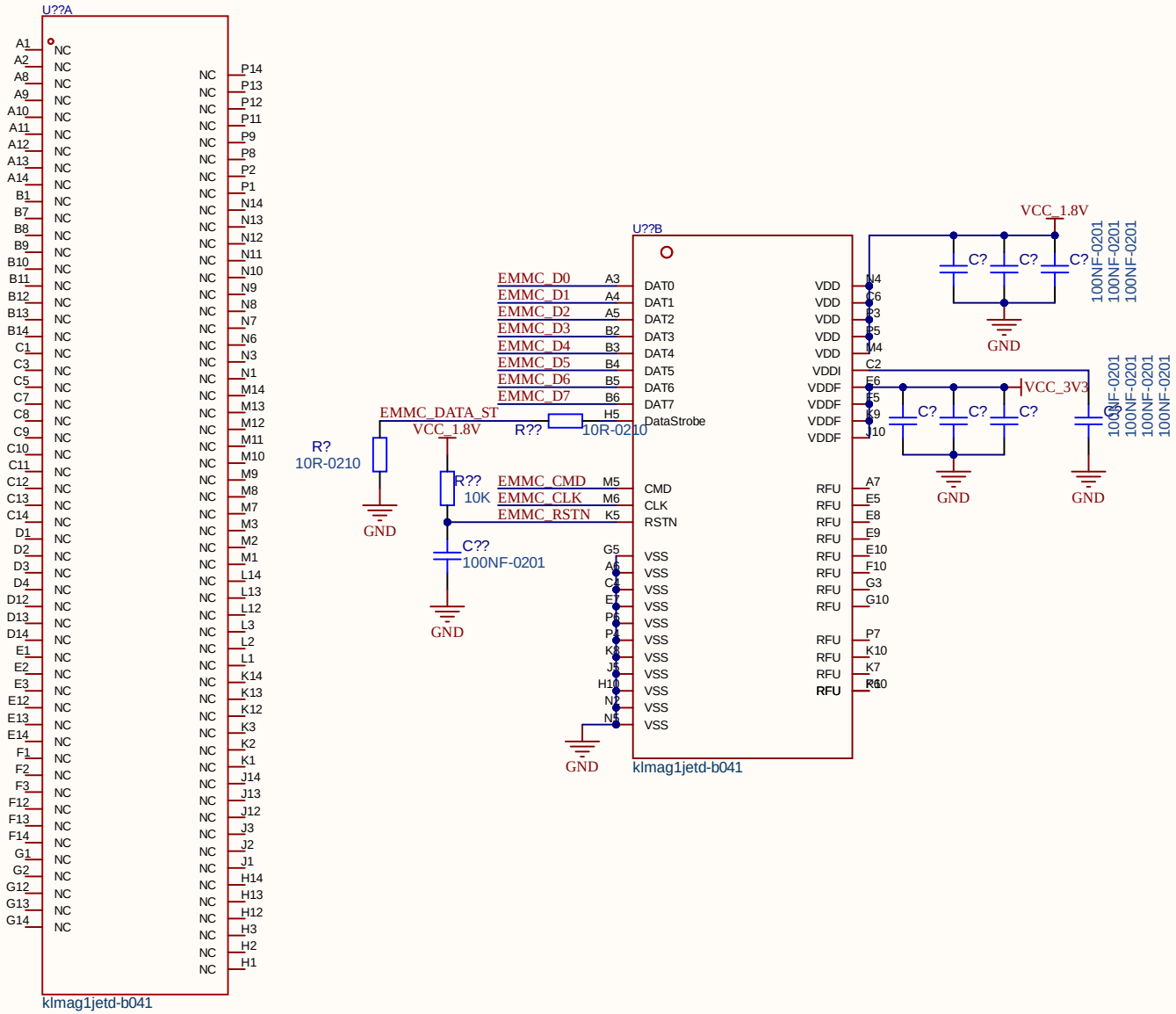


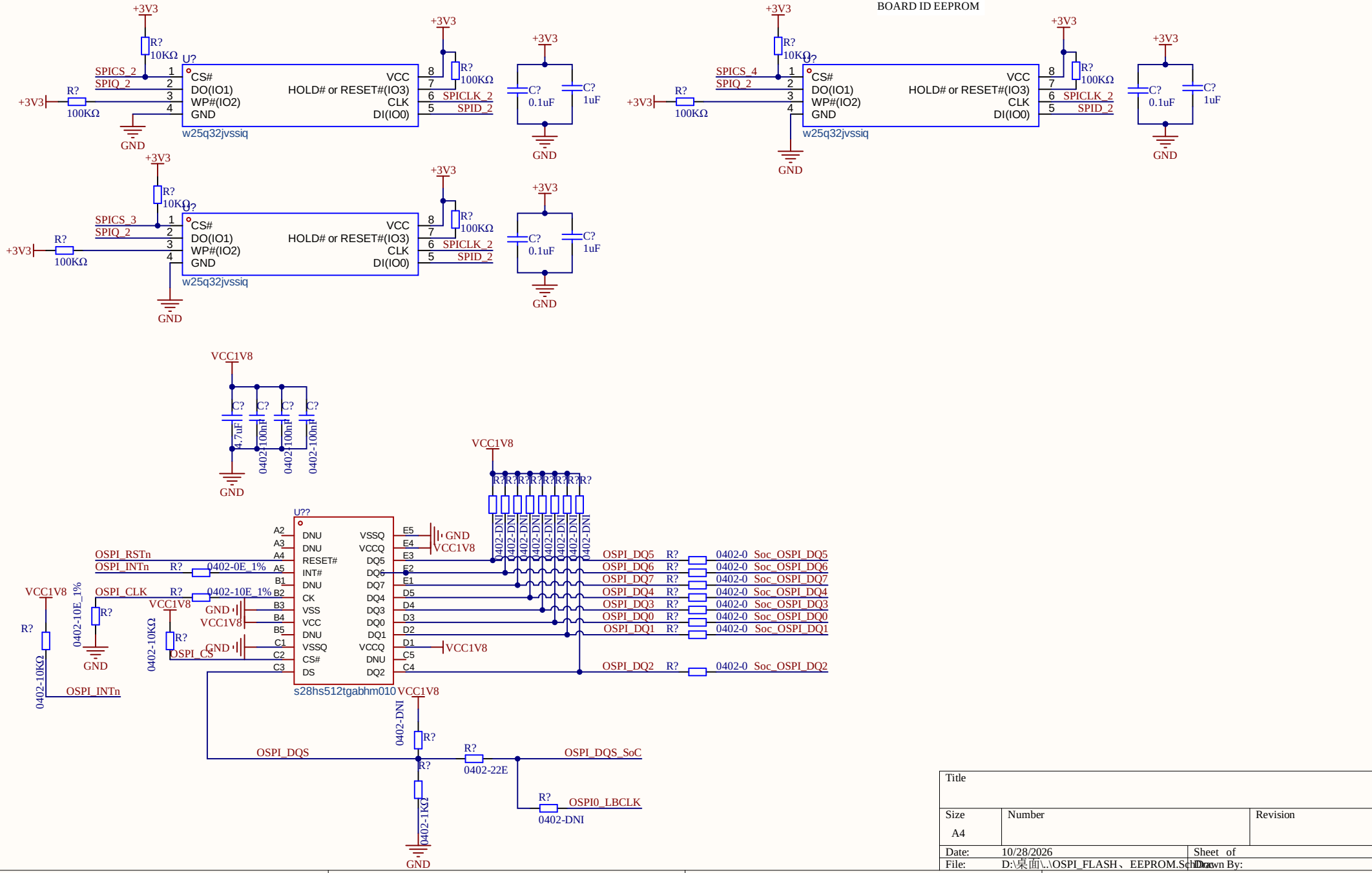

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BOARD ID EEPROM
+3V3
R?
10KΩ
U?
SPICS_2
SPIQ_2
CS#
DO(IO1)
WP#(IO2)
GND
VCC
HOLD# or RESET#(IO3)
CLK
DI(IO0)
SPICLK_2
SPID_2
100KΩ
w25q32jvssiq
C?
0.1uF
1uF
SPICS_4
SPICS_3
VCC1V8
4.7uF
0402-100nF
U??
s28hs512tgabhm010
DNU
RESET#
INT#
CK
VSS
VCC
VSSQ
CS#
DS
VCCQ
DQ5
DQ6
DQ7
DQ4
DQ3
DQ0
DQ1
DQ2
0402-DNI
OSPI_RSTn
OSPI_INTn
OSPI_CLK
OSPI_CS
OSPI_DQS
0402-0E_1%
0402-10E_1%
0402-10KΩ
0402-22E
0402-1KΩ
OSPI_DQS_SoC
OSPI0_LBCLK
OSPI_DQ5
OSPI_DQ6
OSPI_DQ7
OSPI_DQ4
OSPI_DQ3
OSPI_DQ0
OSPI_DQ1
OSPI_DQ2
0402-0 Soc_OSPI_DQ5
0402-0 Soc_OSPI_DQ6
0402-0 Soc_OSPI_DQ7
0402-0 Soc_OSPI_DQ4
0402-0 Soc_OSPI_DQ3
0402-0 Soc_OSPI_DQ0
0402-0 Soc_OSPI_DQ1
0402-0 Soc_OSPI_DQ2
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Size
A4
Number
Revision
Date:
10/28/2026
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# SoC POR_RST




# SoC WARM_RST

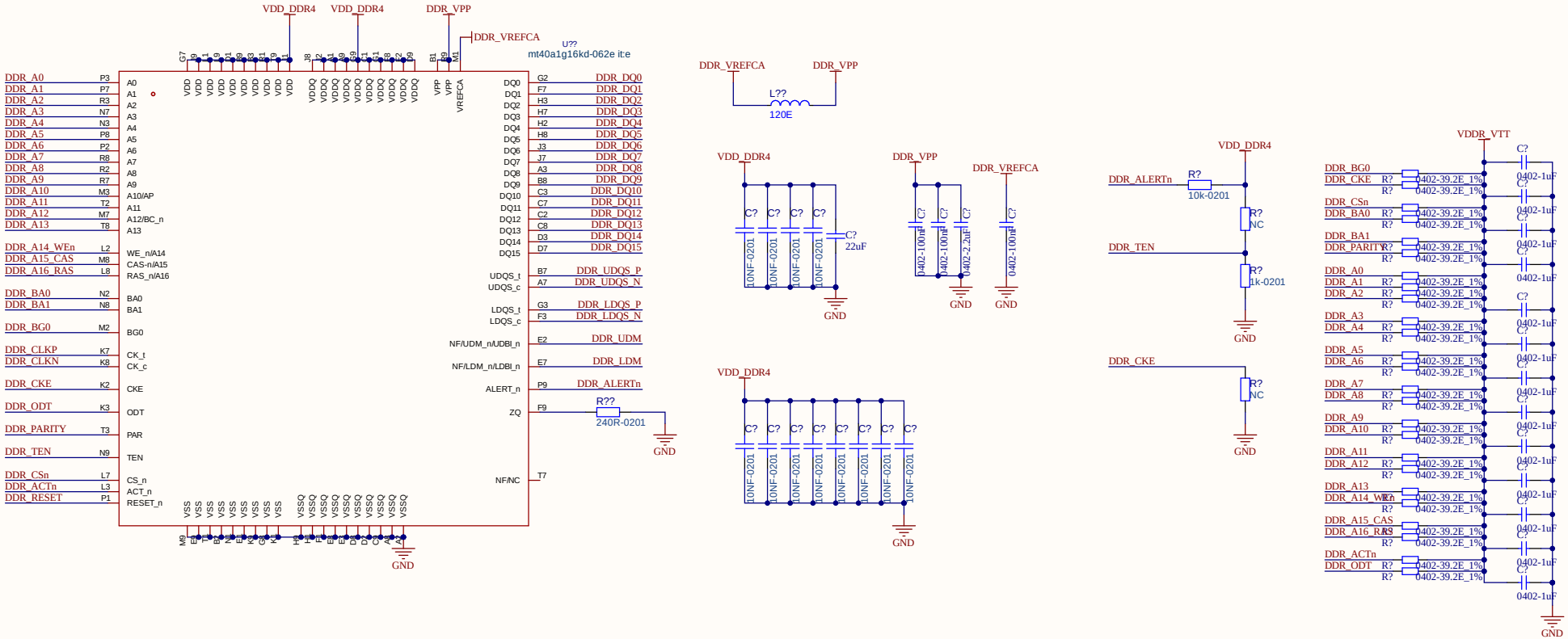




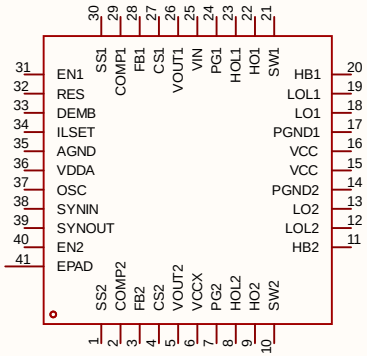


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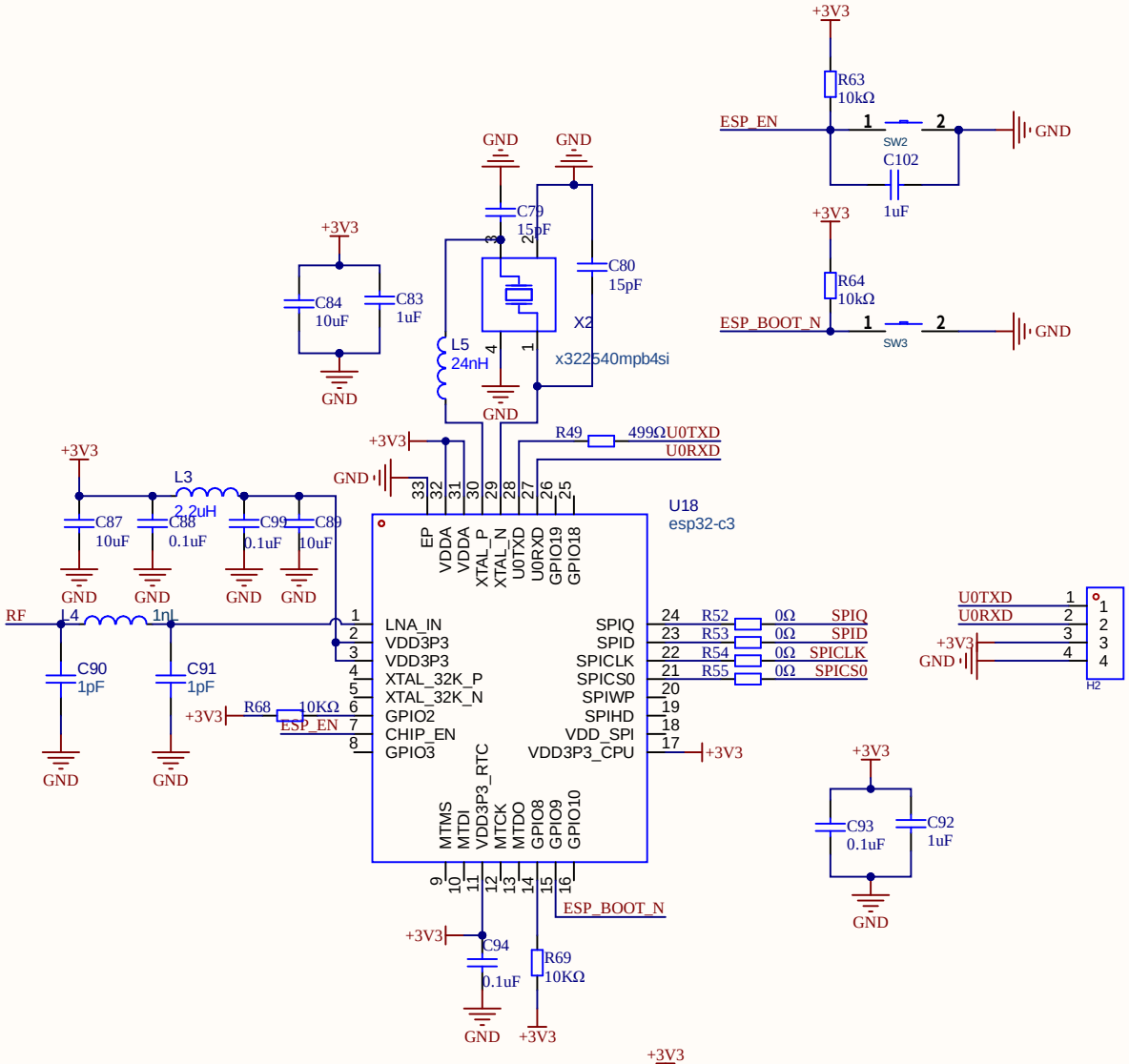
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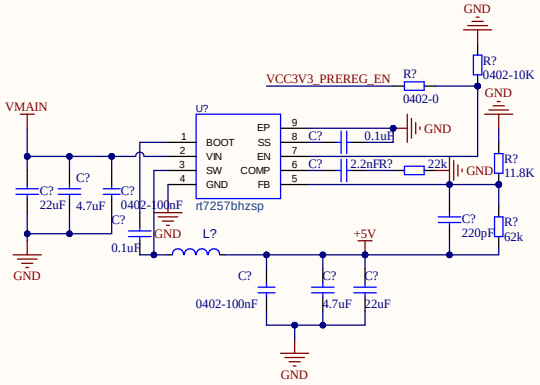



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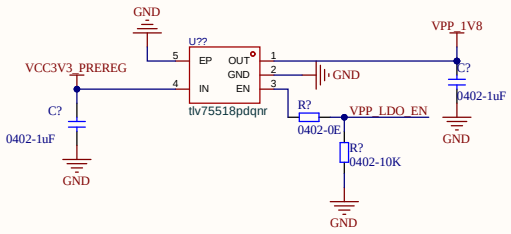



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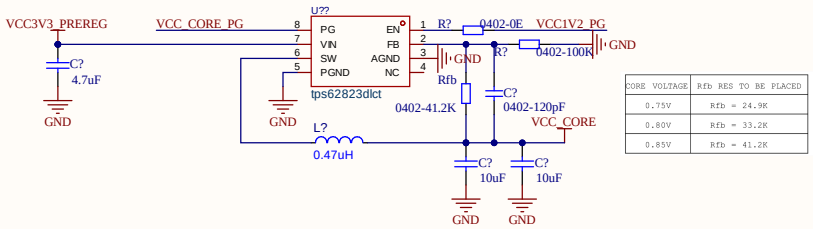
## PREREG 3.3V, 3.0AMPS SUPPLY




## 1.8V VPP, 0.15AMPS SUPPLY

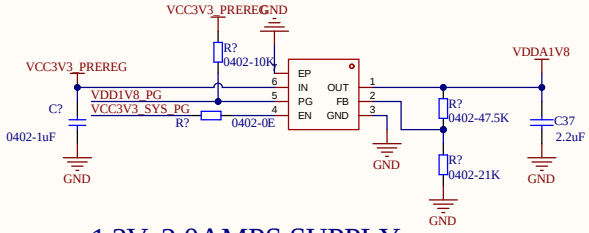



## 0.75 / 0.8 /0.85V, 3.0AMPS SUPPLY

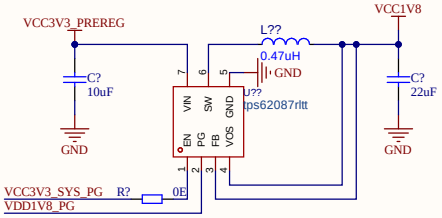



| CORE VOLTAGE | Rfb RES TO BE PLACED |
|---|---|
| 0.75V | Rfb = 24.9K |
| 0.80V | Rfb = 33.2K |
| 0.85V | Rfb = 41.2K |

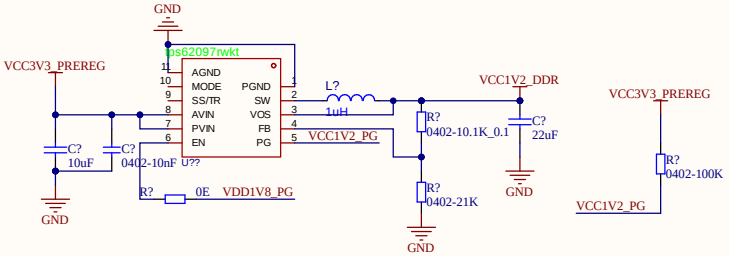
## 1.8V Analog , 1AMPS SUPPLY

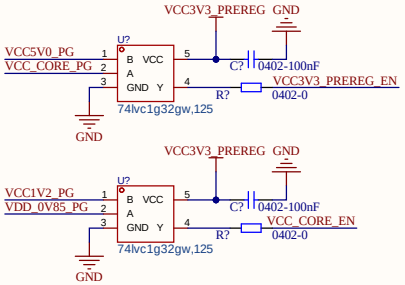



## 1.8V IO, 3.0AMPS SUPPLY



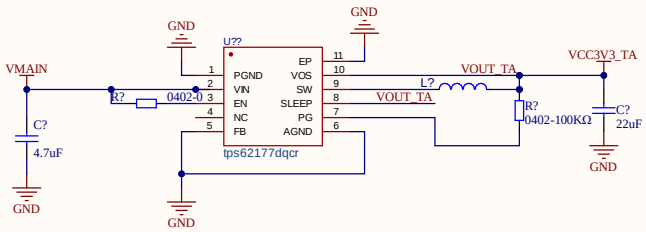

## 1.2V, 2.0AMPS SUPPLY



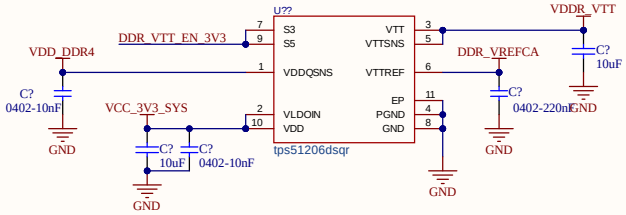




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| Date: 10/28/2026 | Sheet of | |
| File: D:\桌面\..\SoC POWER SUPPLY.SchDoc | Drawn By: | |

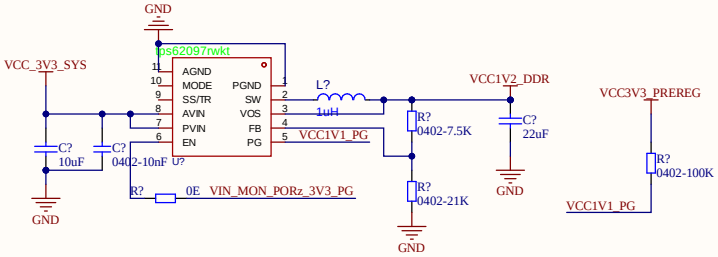
## TEST AUTOMATION BOARD POWER



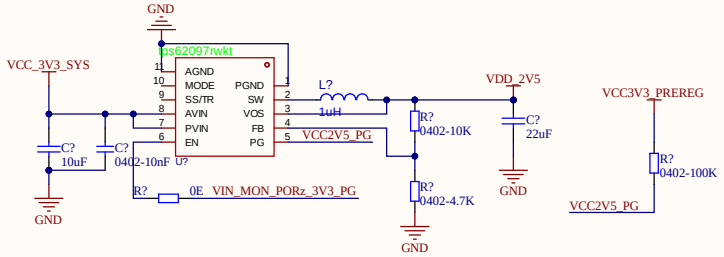

## VTT SUPPLY FOR DDR4



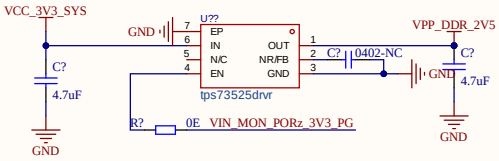

## 1.1V ETHERNET PHY POWER SUPPLY



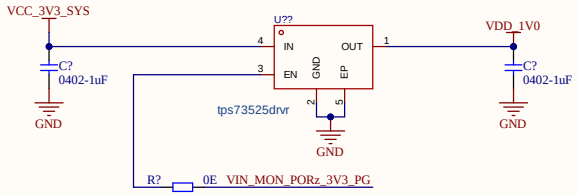

## 2.5V, 2.0AMPS SUPPLY



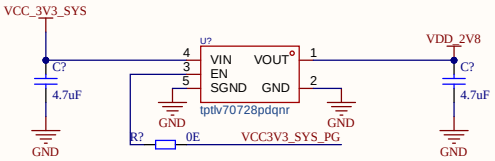

## 2.5V, .5 AMPS SUPPLY



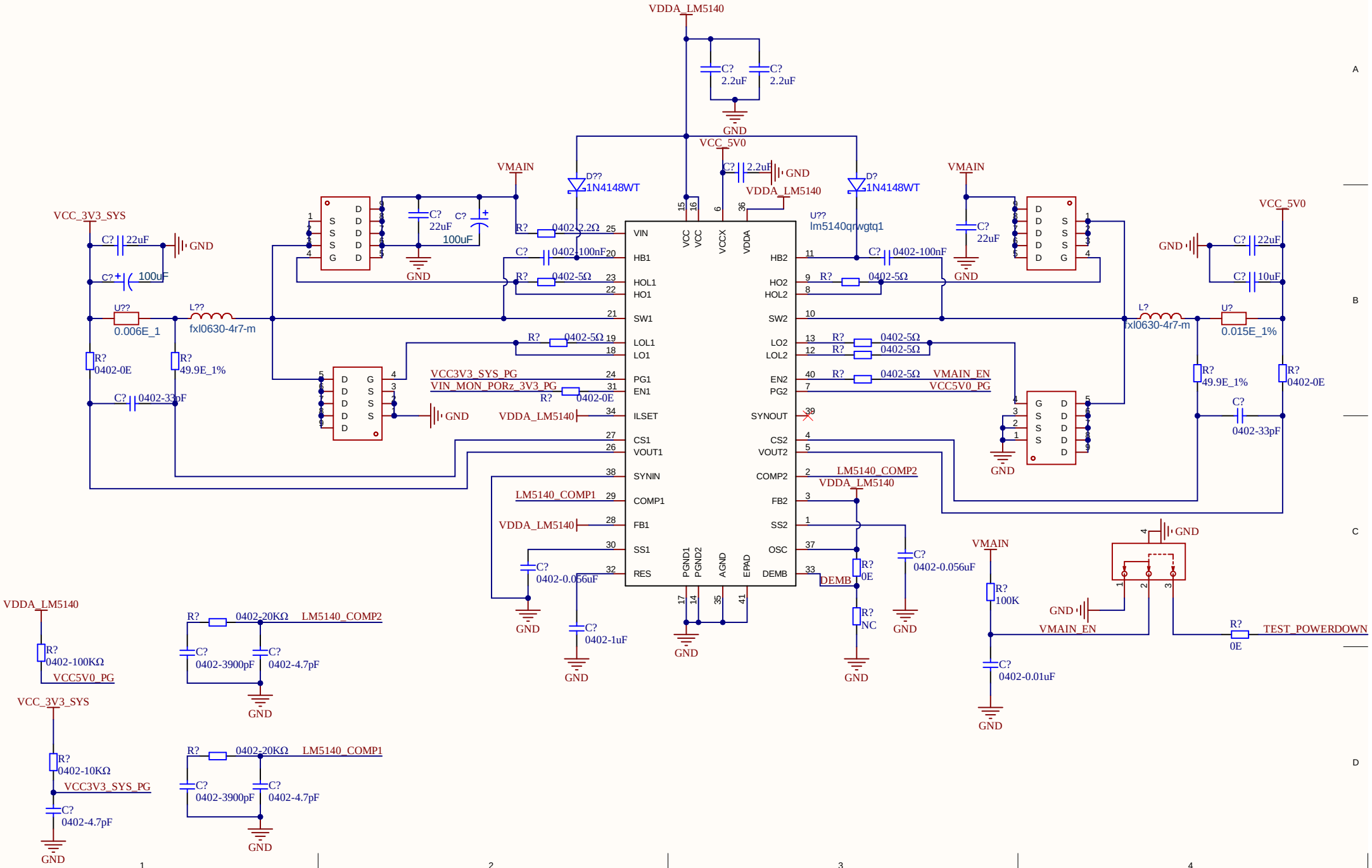

## 1.0V ETHERNET PHY POWER SUPPLY

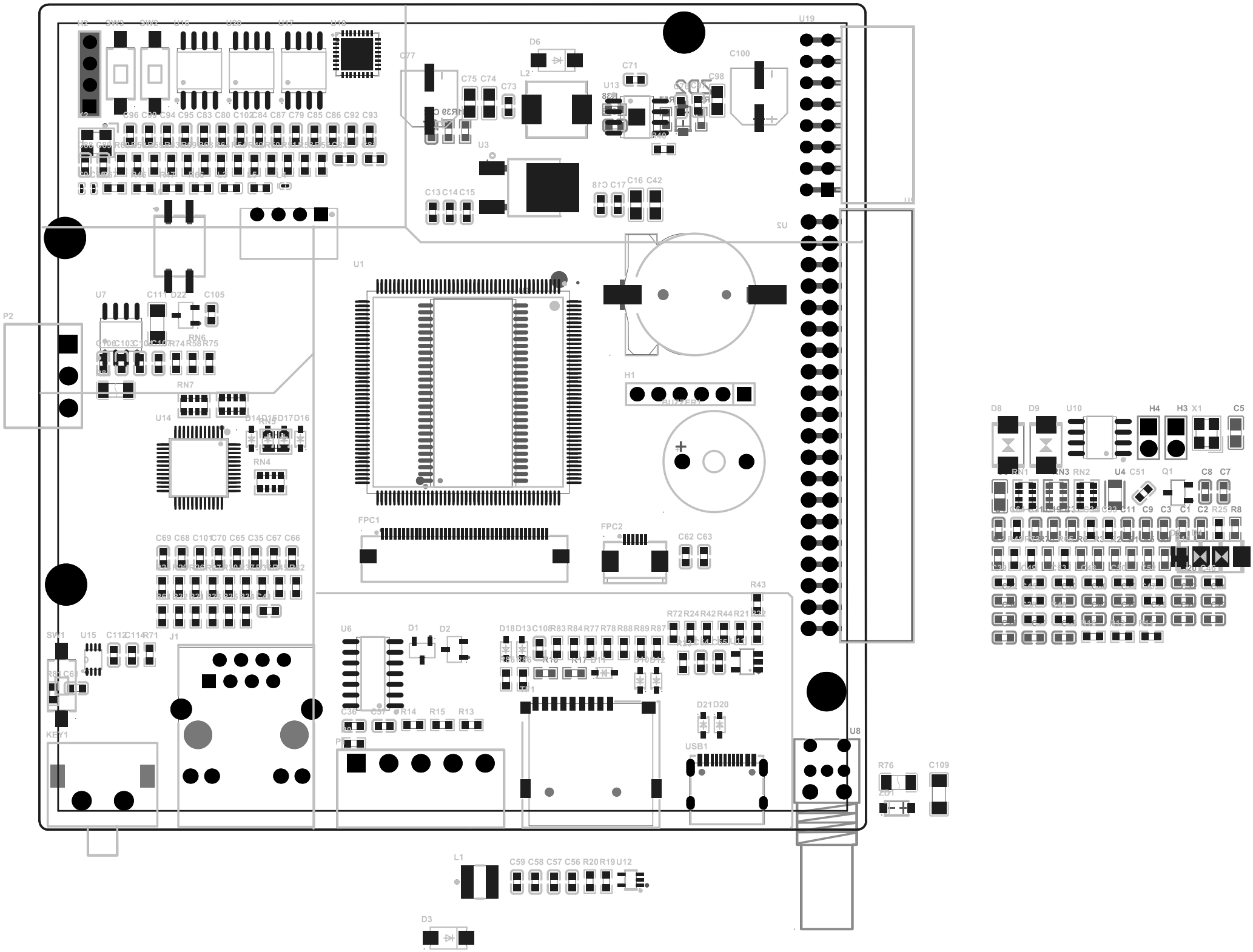



## 2.8V , 0.15AMPS SUPPLY




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VDDA_LM5140
C? 2.2uF
C? 2.2uF
GND
VCC_5V0
VMAIN
D?? 1N4148WT
D? 1N4148WT
U?? lm5140qrwgtq1
VIN
HB1
HOL1
HO1
SW1
LOL1
LO1
PG1
EN1
ILSET
CS1
VOUT1
SYNIN
COMP1
FB1
SS1
RES
VCC
VCCX
VDDA
PGND1
PGND2
AGND
EPAD
HB2
HO2
HOL2
SW2
LO2
LOL2
EN2
PG2
SYNOUT
CS2
VOUT2
COMP2
FB2
SS2
OSC
DEMB
VCC_3V3_SYS
C? 22uF
C? 100uF
U?? 0.006E_1
L?? fxl0630-4r7-m
R? 0402-0E
R? 49.9E_1%
C? 0402-33pF
R? 0402-2.2Ω
C? 0402-100nF
R? 0402-5Ω
VCC3V3_SYS_PG
VIN_MON_PORz_3V3_PG
R? 0402-0E
C? 0402-0.056uF
C? 0402-1uF
LM5140_COMP1
LM5140_COMP2
R? 0E
R? NC
DEMB
VMAIN_EN
VCC5V0_PG
L? fxl0630-4r7-m
U? 0.015E_1%
R? 49.9E_1%
R? 0402-0E
C? 0402-33pF
R? 100K
C? 0402-0.01uF
R? 0E
TEST_POWERDOWN
VDDA_LM5140
R? 0402-100KΩ
VCC5V0_PG
R? 0402-20KΩ LM5140_COMP2
C? 0402-3900pF
C? 0402-4.7pF
VCC_3V3_SYS
R? 0402-10KΩ
VCC3V3_SYS_PG
C? 0402-4.7pF
R? 0402-20KΩ LM5140_COMP1
C? 0402-3900pF
C? 0402-4.7pF
GND


U2
SW3
SW2
U16
U28
U17
U18
U19
D6
C77
C75
C74
C73
L2
C71
U13
C100
C98
U3
C13
C14
C15
C16
C42
C17
U1
U7
C111
D22
C105
P2
RN6
RN7
U14
D14
D15
D17
D16
RN4
H1
FPC1
FPC2
C62
C63
R43
C69
C68
C101
C70
C65
C35
C67
C66
SW1
U15
C112
C114
R71
J1
U6
D1
D2
D18
D13
C108
R83
R84
R77
R78
R88
R89
R87
R72
R24
R42
R44
R21
R22
C36
C37
R14
R15
R13
KEY1
P1
D21
D20
USB1
U8
R76
C109
ZD1
L1
C59
C58
C57
C56
R20
R19
U12
D3
D8
D9
U10
H4
H3
X1
C5
RN1
RN3
RN2
U4
C51
Q1
C8
C7
C11
C9
C3
C1
C2
R25
R8

